# 🏆 Infosys SE Elite Crack Vault — 08 Permutations Combinations and Probability Axioms Elite

🎯 Infosys Systems Engineer (SE) | 📊 16 Questions | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

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## Problem 01 • Permutation, Combination & Probability

🔥 MUST-SOLVE

`ID: SEC02_BOOK_P04_Q21` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> Out of 7 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?

- a. 210
- b. 310
- c. 250
- d. 300

✓ **Verified Answer:** `A`

## Problem 02 • Permutation, Combination & Probability

🔥 MUST-SOLVE

`ID: SEC02_BOOK_P04_Q22` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> In how many ways a committee, consisting of 5 men and 6 women can be formed from 8 men and 10 women?

- a. 11750
- b. 11760
- c. 11520
- d. 16930

✓ **Verified Answer:** `B`

## Problem 03 • Permutation, Combination & Probability

MUST-SOLVE

`ID: SEC02_BOOK_P04_Q23` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?

- a. 756
- b. 500
- c. 699
- d. 321

✓ **Verified Answer:** `A`

## Problem 04 • Permutation, Combination & Probability

MUST-SOLVE

`ID: SEC02_BOOK_P04_Q25` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> In how many ways can 8 Indians and, 4 American and 4 Englishmen can be seated in a row so that all person of the same nationality sit together?

- a. 3! 4! 8! 4!
- b. 3! 8!
- c. 4! 4!
- d. 8! 4! 4!

✓ **Verified Answer:** `A`

## Problem 05 • Permutation, Combination & Probability

MUST-SOLVE

`ID: SEC02_BOOK_P05_Q27` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> A person has to cover 6 meters using either a 1 meter step or a 2 meter step or a combination of both. Find in how many ways you can do it

- a. 10
- b. 12
- c. 25
- d. 13

✓ **Verified Answer:** `D`

## Problem 06 • Permutation, Combination & Probability

MUST-SOLVE

`ID: SEC02_BOOK_P05_Q28` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> Find the number of ways to reach from point A to C if the person has to take the shortest route and he can move in horizontal and vertical direction only

- a. 20
- b. 25
- c. 30
- d. 35

✓ **Verified Answer:** `D`

## Problem 07 • Permutation, Combination & Probability

MUST-SOLVE

`ID: SEC02_BOOK_P08_Q52` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> In how many ways can 7 I. A and 5 I.Sc. Students be seated in a row so that no two of the I.Sc students may sit together?

- a. 7! 8! 3!
- b. 6! 8! 3!
- c. 5! 8! 3!
- d. 4! 8! 3!

✓ **Verified Answer:** `A`

## Problem 08 • Permutation, Combination & Probability

MUST-SOLVE

`ID: SEC02_BOOK_P08_Q54` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> In a birthday party, every person shakes hand with every other person. If there was a total of 28 handshakes in the party, how many persons were present in the party?

- a. 8
- b. 9
- c. 6
- d. 7

✓ **Verified Answer:** `A`

## Problem 09 • Permutation, Combination & Probability

CORE SE

`ID: SEC02_BOOK_P08_Q55` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> There are 10 points in a plane of which 4 are collinear. How many different straight lines can be drawn by joining these points?

- a. 6c2 – 4c2 + 1
- b. 7c2 – 4c2 + 1
- c. 8c2 – 4c2 + 1
- d. 10c2 – 4c2 + 1

✓ **Verified Answer:** `D`

## Problem 10 • Permutation, Combination & Probability

CORE SE

`ID: SEC02_BOOK_P08_Q56` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> There are 10 points in a plane out of which 4 are collinear. Find the number of triangles formed with these points as vertices?

- a. 10c3 - 4c3
- b. 12c3 - 4c3
- c. 14c3 - 4c3
- d. 15c3 - 4c3

✓ **Verified Answer:** `A`

## Problem 11 • Permutation, Combination & Probability

CORE SE

`ID: SEC02_BOOK_P13_Q95` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> In a simultaneous throw of two dice find the probability of getting a total of 8.

- a. 5 36
- b. 4 36
- c. 3 36
- d. 2 36

✓ **Verified Answer:** `A`

## Problem 12 • Permutation, Combination & Probability

CORE SE

ID: SEC02_BOOK_P13_Q96 | Source: 250 + Video Solution Quant Assorted Book.pdf | Level: Infosys SE Standard

> Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?

- a. 3 20
- b. 4 20
- c. 6 20
- d. 9 20

✓ **Verified Answer:** `D`

## Problem 13 • Permutation, Combination & Probability

CORE SE

ID: SEC02_BOOK_P13_Q97 | Source: 250 + Video Solution Quant Assorted Book.pdf | Level: Infosys SE Standard

> Two dice are thrown simultaneously. What is the probability of getting two numbers whose product is even?

- a. 1 4
- b. 2 4
- c. 3 4
- d. 4 4

✓ **Verified Answer:** `C`

## Problem 14 • Permutation, Combination & Probability

CORE SE

ID: SEC02_BOOK_P14_Q100 | Source: 250 + Video Solution Quant Assorted Book.pdf | Level: Infosys SE Standard

> A basket contains 5 red 4 blue 3 green marbles. If three marbles picked up random, what is the probability that either all are green or all are red?

- a. 1/20
- b. 7/20
- c. 3/20
- d. 9/20

✓ **Verified Answer:** `A`

## Problem 15 • Permutation, Combination & Probability

CORE SE

ID: SEC02_BOOK_P14_Q101 | Source: 250 + Video Solution Quant Assorted Book.pdf | Level: Infosys SE Standard

> A bag contains 5 red caps, 4 blue caps, 3 yellow caps and 2 green caps. If three caps are picked at random, what is the probability that two are red and one is green?

- a. 22/55
- b. 15/81
- c. 10/91
- d. 5/91

✓ **Verified Answer:** `D`

## Problem 16 • Permutation, Combination & Probability

CORE SE

`ID: SEC02_BOOK_P14_Q103` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> Out of 15 students studying in a class, 7 are from Maharashtra, 5 are from Karnataka and 3 are from Goa. Four students are to be selected at random. What are the chances that at least one is from Karnataka?

- a. 11 13
- b. 12 13
- c. 13 13
- d. 14 13

✓ **Verified Answer:** `A`